COSC2148/COSC3130 Computing Research and Project Preparation

Assignment 2

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Section 1

Introduction

Widespread misinformation, generating challenges for information credibility alongside informed decision-making can be attributed to the rise of social media. The increasing complexity of online content consisting of textual, visual, and interaction-based elements, has increased the difficulty of accurately identifying false information. Thus, the development of reliable misinformation detection systems has become a critical research focus.

Early research in misinformation detection has been predominantly characterised using text-based classification models, where linguistic features are utilised to determine content authenticity. However, limitations in these approaches have been observed, predominantly in the inability to capture non-textual signals embedded within modern social media content. Multimodal learning has partially overcome this limitation by integrating text alongside images. For example, EANN framework demonstrated that combining both text and images leads to enhanced detection performance compared to unimodal approaches (Wang et al., 2018). Similar to this, large-scale multimodal dataset such as Fakeddit allows better testing, showing that multimodal representations provide superior accuracy when detecting complex misinformation classes (Nakamura et al., 2020). Moreover, recent progress in multimodal architectures has been made towards cross-modal interaction and multimodal feature fusion.Models such as HMCAN and MPFN prove that modelling relationships between text and image modalities significantly enhances detection performance (Qian et al., 2021; Jing et al., 2023).

In parallel, the incorporation of social context information has been explored through the development of context-aware detection models. These approaches utilise features derived from user interactions, propagation patterns, and temporal dynamics to improve classification performance. For example, models using graph representations of temporally evolving graphs to represent social media dynamics allow for the modelling of information diffusion in the social media environment (Song et al., 2021). On the other hand, models based on social interactions between posts and social media users have shown the ability of social interactions to be employed to detect misleading posts (Min et al., 2022). Moreover, recent advances have demonstrated that incorporating both modalities of content and social context data enhances the performance of the classifier (Nadeem et al., 2023; Jing et al., 2025).

Despite these advancements, a crucial limitation remains being the lack of systematic integration between multimodal content modelling and context-aware architectures. Existing context-aware frameworks (i.e. DANES and GETAE), retain a text-only content branch restricting their ability to fully leverage multimodal signals present in contemporary social media environments (Truică et al., 2024).Whilst prior studies have explored the combination of multimodal content and social context, limited research has conducted controlled evaluations that isolate the contribution of each component within a unified framework (Dellys et al., 2025). This results in an incomplete understanding of the relative impact of multimodal content features and social context features on detection performance.

This research proposes the development of a context-aware multimodal misinformation detection framework, in which the traditional text-based content branch is substituted with a multimodal content branch that integrates both textual and visual representations. The research adopts a comparative experimental design to enable the systematic evaluation of architectural modifications through controlled substitution and ablation-based analysis. The objective of this research is to determine whether the integration of multimodal features within a context-aware framework results in measurable performance improvements, and to analyse the relative contribution of multimodal content and social context components to overall detection effectiveness.

The contribution of this research is threefold. First, existing context-aware architectures are extended through the incorporation of a multimodal content branch. Second, a structured experimental framework is introduced to enable direct attribution of performance improvements through controlled ablation. Third, empirical insights are generated into the interaction between multimodal content features and social context signals. The significance of this research lies in its potential to support the development of more robust and contextually aligned misinformation detection systems that reflect the complexity of real-world social media environments.

The structure of this paper is organised as follows. In Section 2, the research questions are presented, which provide a basis for the problems to be solved through this study. Section 3 is concerned with an analysis of previous research, particularly on multimodal and context-aware systems of misinformation detection, highlighting their limitations. The research methodology, comprising details on system architecture, experiment setup, and evaluation method, is described in Section 4. Section 5 deals with describing the procedure for evaluating the performance of our model against previous benchmark models. Finally, Section 6 explains the approach adopted for project management.

Section 2

Research Questions

2.1 Research Questions

Research questions - 1 page (4 marks)

This section should address the technical issues you will be addressing in your research. It is about the "what" (you will be addressing/focussing when you start your thesis).

Describe here the specific research questions to be addressed. You need to break the problem you are addressing into specific sub-problems (called here research questions) that could help you during the thesis to come up with innovative solutions properly.

Show how the research questions are interesting and worth addressing.

Please include at least two (2) research questions, but you are limited to a maximum of three (3) research questions.

For each research question, describe what it is, its significance, and why it is important to address it.

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2.3 Research Questions

2.3.1 Research Questions

Overview

- This research focuses on evaluating the integration of multimodal content within context-aware fake news detection models.
- The study aims to investigate how combining multimodal features with social context affects detection performance.

- The research questions are designed to break the problem into key sub-problems related to model performance and component contribution.

RQ1: Effect of Multimodal Content Integration

- How does replacing a text-only content branch with a multimodal content branch affect the performance of context-aware inauthentic social media content detection models, and to what extent?
- Significance:
  - Evaluates whether multimodal representations improve detection accuracy.
  - Provides direct comparison with existing context-aware models such as DANES and GETAE.
  - Helps quantify the performance gain from incorporating visual features alongside text.
- Why it is important:
  - Addresses a key limitation in current models that rely on text-only content.
  - Helps determine if multimodal integration is worth the added complexity.

RQ2: Contribution of Model Components

- How does the relative contribution of the multimodal content branch and the social content branch affect overall detection performance?
- Significance:
  - Analyses the importance of different components within the model.
  - Identifies whether content features or social context features contribute more to performance.
- Why it is important:
  - Supports better design of hybrid detection frameworks.
  - Helps prioritise which components should be optimised in future models.

Section 3

Literature Review

3.1 Overview of Social Media Misinformation Detection

The development of misinformation detection in social media can be observed across three main directions: text-based content analysis, social-context modelling, and multimodal detection. Earlier studies framed misinformation detection as a content classification problem, focusing on the linguistic and semantic properties of posts. While these approaches established useful baselines, increasing limitations have been identified in modern social media environments, where misleading information is conveyed not only through text but also through images, memes, screenshots, and user interaction patterns. As a result, a shift toward multimodal learning and context-aware modelling has been observed, reflecting the need to capture broader signals available within online platforms (Wang et al., 2018; Qian et al., 2021; Truică et al., 2024; Truică et al., 2025).

3.2 Text-Only Content-Based Detection Approaches

Text-only content modelling has remained a foundational component in misinformation detection, including within later context-aware systems. In the relevant literature, post content continues to be represented primarily through text-based features. The main strength of this approach is the provision of a strong semantic baseline, particularly when misinformation is expressed through written language. However, limitations are increasingly evident in social media contexts, where misinformation is often conveyed through visual and multimodal formats. This is visible in models such as DANES and GETAE, where text-only representations are still used within the content branch despite strong performance (Truică et al., 2024; Truică et al., 2025). Thus, while useful as a baseline, text-only modelling provides a restricted view of content and is insufficient for capturing the full complexity of contemporary inauthentic posts.

3.3 Context-Aware Detection with Social and Propagation Signals

A major research direction involves the incorporation of social context into misinformation detection. Rather than focusing only on post content, these approaches model information spread, user engagement, and temporal interaction patterns. Evidence shows that diffusion patterns, user relationships, and temporal dynamics provide strong classification signals, often capturing behavioural evidence unavailable from content alone (Song et al., 2021; Min et al., 2022; Zhu et al., 2024). Models such as DANES and GETAE represent important context-aware ensemble architectures, combining a content branch with a social or propagation branch. Their strength lies in structured use of social context and strong benchmark performance. However, reliance on a text-only content branch limits the ability to capture visual misinformation. Additionally, propagation-based methods depend on interaction data that may be sparse or delayed, particularly in early detection scenarios. Consequently, while detection extends beyond content, limitations of text-only representation are still inherited (Truică et al., 2024; Truică et al., 2025).

3.4 Multimodal Content-Based Detection Approaches

Multimodal detection addresses the limitations of text-only modelling. Foundational work such as EANN demonstrated improved performance through combining textual and visual features (Wang et al., 2018). Later studies focused on improving fusion methods. For example, MCNN models text-image consistency, while HMCAN and MPFN show that hierarchical attention and progressive fusion better capture cross-modal relationships than simple concatenation (Xue et al., 2021; Qian et al., 2021; Jing et al., 2023). This progression indicates that gains depend not only on modality inclusion but also on interaction quality. Benchmark datasets such as Fakeddit further show that multimodal classifiers outperform unimodal baselines (Nakamura et al., 2020; Segura-Bedmar & Alonso-Bartolome, 2022). While this literature reflects modern social media content well, a key limitation is the focus on content-level modelling, with limited integration of social and propagation context.

3.5 Integrating Multimodal Content and Social Context

Recent research has explored integrating multimodal content and social context within unified frameworks. Studies such as EFND, C^2DCMTN, DPSG, TMEF-BI, and ESDGFN show that combining textual, visual, and contextual signals can achieve strong performance (Nadeem et al., 2023; Wang et al., 2024; Jing et al., 2025; Chen et al., 2025; Sun et al., 2025; Ji et al., 2026). These findings indicate that multimodal content and social context are complementary. However, variation in how social context is represented introduces challenges in comparison. Some models use engagement data, while others incorporate behaviour, stance, or dynamic graph fusion (Nadeem et al., 2023; Wang et al., 2024; Chen et al., 2025; Sun et al., 2025). This lack of consistency prevents clear attribution of performance gains, as both multimodal and social components are often changed simultaneously. As a result, while joint modelling is supported, the specific contribution of multimodality remains unclear, particularly compared to DANES and GETAE architectures (Truică et al., 2024; Truică et al., 2025; Dellys et al., 2025).

3.6 Research Gap and Positioning of the Proposed Study

A central gap in the literature is identified. Prior research shows that both multimodal content and social context improve misinformation detection (Wang et al., 2018; Qian et al., 2021; Truică et al., 2024; Nadeem et al., 2023; Jing et al., 2025). However, limited research performs controlled substitution of the text-only content branch within established context-aware ensembles while keeping the social branch unchanged. As a result, the specific contribution of multimodal substitution within DANES/GETAE remains unclear, and the relative contribution of content and social context is not well decomposed (Truică et al., 2024; Truică et al., 2025; Dellys et al., 2025; Chen et al., 2025). This gap aligns with the project's research questions. Research Question 1 evaluates performance improvement from multimodal substitution, while Research Question 2 examines whether multimodal and social branches provide complementary or overlapping value. To address this, the study uses DANES and GETAE as baselines, replaces the text-only content branch with a multimodal alternative, and conducts ablation-based comparisons across configurations. This enables a controlled evaluation of the contribution of multimodal content and social context in misinformation detection.

Section 4

Research Methodologies

This research adopts a comparative experimental design to investigate the effectiveness of multimodal content and social context features in the detection of inauthentic social media posts.

note Dellys et al. (2025) is flagged as the closest prior work (how does this differentiate?)

4.1 System Overview

Prior research has demonstrated that combining text-based content features with social context improved detection performance in identifying inauthentic social media posts. Separate studies have shown that multimodal models can outperform text-based models.

Limited research has been conducted to investigate the combined effectiveness of multimodal content features and social context features by substituting text-based representations with multimodal representations within a context-aware framework.

The proposed system aims to address this gap by utilising existing DANES and GETAE feature ensemble architectures that combine text content with social context features, and extending them to incorporate multimodal content features.

4.2 Proposed Framework Architecture

4.2.1 Baseline Architecture: DANES/GETAE

DANES (citE) and GETAE (cite) each comprise two branches: a text-based content branch that encodes textual content using word embeddings and transformer-based representations in a neural network, and a social context branch (referred to as the Propagation Branch in GETAE) that models the social network user interactions and behaviour using graph neural networks and deriving node embeddings.

Outputs from these branches are combined using an ensemble approach produce a classification embedding. These models achieve state-of-the-art performance in detecting inauthentic social media posts in Twitter15 and Twitter16 datasets, outperforming text-only models and other context-aware models that do not utilise multimodal content features.

These models represent a principled starting point for the proposed architecture modification.

4.2.2 Proposed Modifications to Baseline Architecture

The proposed architecture modification involves substituting the text-based content branch with a multimodal content branch that incorporates visual features extracted from social media posts as well as textual features. The social context branch is retained, as in the baseline architectures, allowing the contribution of the multimodal content branch to be evaluated against the baseline models.

4.2.3 Multimodal Branch Design

This research will adopt a well validated multimodal architecture from the literature that has been designed for the task of multimodal fake news detection. This approach maintains a focus on the architectural substitution question rather than the design and optimisation of a novel multimodal architecture.

The key considerations for the selection are: task alignment, dataset compatibility, ease of integration and widespread validation.

From our review of the literature, HMCAN and MPFN models have both demonstrated excellent accuracy (0.89.7% and 83.3% respectively) in multimodal fake news detection using the Multimodal Twitter dataset, outperforming other common multimodal models for fake news detection (such as EANN). HMCAN's BERT + ResNet encoding pipeline is consistent with the text branch of DANES/GETAE and has been independently validated on the Multimodal Twitter dataset as well as Fakedit dataset, making it a suitable candidate for the multimodal content branch. MPFN utilises a more complex image pipeline which could be beneficial for detecting manipulated or artificially generated images, and is another potential candidate.

Our system will adopt HMCAN with MPFN as a fallback option as this provides a robust foundation for handling multimodal content.

4.2.4 Experimental Design

The initial stage is to replicate the published DANES/GETAE architectures and evaluate their performance on the Twitter15 and Twitter16 datasets and ensure that the results are consistent with those reported in the original papers.

This is followed by the transition to a dataset that contains multimodal content and the implementation of the proposed architecture modification to incorporate multimodal content features.

The following configurations are proposed for evaluation:

- **Stage 1 — Condition A (Implementation Baseline):** Replicate the published DANES/GETAE architectures using the Twitter 15/16 datasets to confirm the implementation reproduces performance consistent with the original papers before any modifications are introduced.
- **Stage 2 — Condition A (Dataset Baseline):** Evaluate the DANES/GETAE architectures on a multimodal dataset in text-only mode (i.e. ignoring the visual content)

to establish a baseline for the DANES/GETAE architectures on the new dataset that Stage 2 conditions will be compared with.

- **Stage 2 — Condition B (Ablation 1):** Text-only content branch with no social context branch to isolate the contribution of textual features without social context features.
- **Stage 2 — Condition C (Ablation 2):** Multimodal content branch with no social context branch to isolate the contribution of multimodal features without social context features.
- **Stage 2 — Condition D (Proposed):** Multimodal content branch with a social context branch to represent the full proposed framework for primary evaluation.

Each stage 2 condition is compared against the stage 2 condition A baseline to evaluate the attribution of performance improvements.

RQ1 is directly addressed by comparing the performance of the proposed architecture (Condition D) against the baseline architecture (Condition A) with the multimodal dataset.

A pairwise comparison of condition's B, C and D address RQ2 by evaluating the contribution of the multimodal content branch and the social context branch to the overall performance of the model.

It is anticipated that the transition from Twitter15/16 datasets to a multimodal datasets will result in a performance difference in Stage 2 Condition A relative to Stage 1 depending on the extent to which these models generalise to a new dataset. This does not impact the evaluation of the proposed architecture change since all Stage 2 conditions are evaluated against the Stage 2 Condition A baseline. (TODO: this would require a mention under the LIMITATIONS section, we aren't interested in researching the generalisability of these models this is rather an expected degradation)

This methodology enables direct evaluation of the research questions by isolating the impact of multimodal content through controlled substitution of the content branch and by analysing the relative contribution of content and context features through ablation experiments. The structured experimental design ensures that performance differences can be attributed to specific components of the model.

4.2.5 Processing Pipeline

4.2.6 Evaluation Metrics

4.2.7 Datasets and Data Preparation

4.2.8 Timeline

Section 5

Evaluation

5.1 DELETE original question

evaluation - 1 page (3 marks)

This section is important and should clearly show the evaluation plan of your research. This evaluation plan could be used later during your thesis to clearly demonstrate that you made research contributions in the field of study.

This section will need to address the following points:

- Benchmarking: clearly select some existing methods/algorithms you will need to compare your research work with. It is important to clearly explain these methods/algorithms as well as justify the reasons for choosing them.
- Data sets: describe at least two well-accepted data sets that you will be using to compare your research work against the selected benchmarked methods/algorithms. Again, you will need to justify the reasons for selecting such data sets.
- Metrics: to compare your research work with benchmarked methods/algorithms, you clearly need to list the metrics to be used for this purpose. This could be, for example, "execution time" if you are comparing the performance of your work with other methods/algorithms. You also need to explain why you have selected such metrics. List at least two metrics.

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Section 6

Project Management

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project management - 1 page (1 mark)

This section should outline how you plan to manage your research project throughout the semester.

It is essentially a Project Plan that clearly defines the scope and deliverables, including explicit exclusions.

It should present a structured schedule with milestones and a transparent critical path, while assigning responsibility to specific 2individuals to ensure accountability.

It must also incorporate risk management to anticipate threats and formulate appropriate responses, protecting project objectives from avoidable failure.

This section should also clearly state the methods you will use to manage the project, such as Agile sprints, Scrum ceremonies, Kanban, etc., and specify the enabling tools (e.g., MS Project, Trello, JIRA, Slack, etc.) that will operationalise these methods.

Briefly explain how these methods and the supporting tools together will facilitate scheduling, progress tracking, communication, and reporting. Specifically, you must include the following:

- High-level scope and deliverables of your research project. Clearly state exactly what you will deliver and explicitly rule out what you will not.
- Timeline with major milestones derived from the Work Breakdown Structure (WBS). This should reflect the schedule of your research project and include any critical paths. Note: you don't need to include the WBS itself.
- Clear assignment (ownership) of each task to project team member(s), not roles, for accountability and monitoring.
- Approach in risk management, identify risks and proposed mitigation and contingency plan, as well as escalation paths. List the major risks, rank them, and state what you will do before, when, and after they occur.

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Bibliography

Appendix A

Generative AI Usage

Declaration

[REPLACE THIS LINE WITH YOUR GENERATED DECLARATION FROM `https://aiattribution.github.io/`]

Tool Report: [Tool Name, e.g. ChatGPT / RMIT Val / Copilot]

Field	Details
1. GenAI tool used	[e.g. ChatGPT 4o, RMIT Val, GitHub Copilot]
2. Purpose for this assignment	[e.g. Brainstorming research questions, drafting outline, rewriting for clarity, code generation]
3. Prompts or instructions given	[Include key prompts used. If conversation was lengthy, include all your prompts in full and use [...] to truncate AI responses only.]
4. Number of prompts given	[e.g. 7]
5. Summary of GenAI output	[Describe what the AI produced. Include brief excerpts where relevant.]
6. Verification of GenAI output	[Explain how you checked the AI's output for factual accuracy, source validity, and logical soundness. e.g. "All cited sources were manually located and verified via Google Scholar."]
7. Usage or modification of output	[Explain how you used, adapted, or discarded the AI-generated content in your final submission.]
8. Reflection	[What worked well? What did not? How did it support your learning? What value did it add?]

Table A.1: Generative AI usage report — [Tool Name]